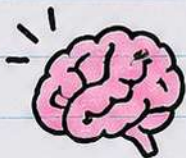


⚡ Linux

Process Management

≡ Day 14/20 ≡



Learn how Linux controls
every running program.



Save this guide



Swipe to learn



2. What is a Process?



★ A **process** is a program that is currently running in the system.

→ Examples:

- Google Chrome 
- VS Code 
- Terminal 
- Firefox 

 Every running process has a unique **PID** (Process ID).

Linux can run many processes at the same time. 

 **FOLLOW**
@devopswithanu

Swipe →

3. ps Command



★ ps (process status) command shows running processes.

Basic Usage :

ps

More Details :

ps -ef

It shows :

- PID (Process ID)
- User
- CPU usage
- Memory usage
- Command

UID	PID	PPID	C	STIME	CMD
anu	1234	1	0	10:15	bash
anu	2345	1234	0	10:16	code
root	3456	1	0	10:17	sshd



PID is a unique number given to every running process.



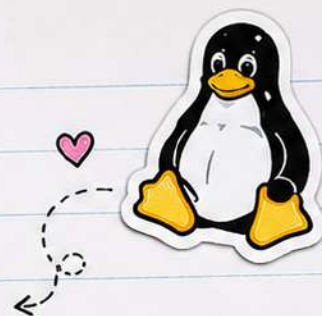
FOLLOW
@devopswithanu

Swipe →





4. top Command



★ top command gives a real-time live view of system processes.

Syntax :

`top`

It shows :

- CPU usage
- Memory usage
- Running processes
- Load average
- Uptime

```
top - 10:27:15 up 1:23, 2 users,
Tasks: 148 total, 1 running, 147 sleeping
%Cpu(s): 3.1 us, 1.2 sy, 0.0 ni,
Mem: 7.7G total, 2.1G used, 5.6G free
```

PID	USER	%CPU	%MEM	TIME+	COMMAND
1234	anu	2.1	1.0	0:02.31	bash
2345	root	1.3	0.6	0:01.15	sshd
3456	anu	0.7	0.3	0:00.45	code
...					



TIP : Press `q` to quit the top view.
Press `h` for help.



FOLLOW
@devopswithanu

Follow for more Linux
tips and tricks ♥

Swipe →



5. htop



Better Process Viewer

★ htop is an interactive process viewer.
It is more powerful than top.



htop



Features :

- ✓ Colorful interface
- ✓ Easy navigation
- ✓ Search processes
- ✓ Kill processes

```

CPU [|||||] 12.5%
Mem [|||||] 2.1G/7.7G
Swp [ ] 0.0G/2.0G

```

PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%	TIME+	Command
1234	anu	20	0	245M	52100	23000	S	2.5	0.7	0:12.34	bash
2345	root	20	0	512M	120M	45000	S	1.8	1.5	0:08.21	sshd
3456	anu	20	0	1.2G	220M	65000	S	1.2	2.8	0:05.12	code
4567	anu	20	0	980M	156M	52000	S	0.8	2.0	0:03.45	chrome
...											

F1 Help

F2 Setup

F3 Search

F4 Filter

F5 Tree

F6 Sort

F9 Kill

F10 Quit

Install :



\$ sudo apt install htop

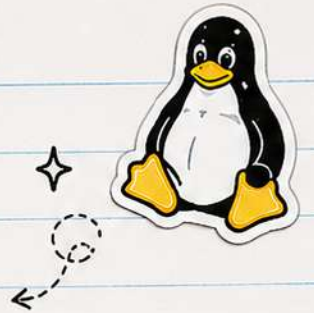
Makes process monitoring much easier! 😊



Swipe →



6. kill Command



★ Used to **terminate** (stop) processes.

Syntax :

`kill [options] <PID>`

Common Options :

- `kill <PID>` → Send default termination signal
- `kill -9 <PID>` → Forcefully kill a process



Example :

- `kill 1234` → Stop process with PID 1234
- `kill -9 1234` → Force kill process with PID 1234
- `killall firefox` → Kill all processes named firefox



Use with care.
You can stop important
processes too! ❤️



Day 14/20

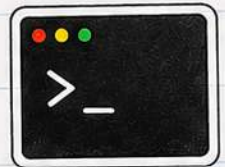
7. nice Command

Adjust the priority of processes



The nice command starts a process with a modified scheduling priority (**niceness**).

Higher nice value → Lower priority



Syntax :

```
nice -n <value> <command>
```

Examples :



```
nice -n 10  
firefox
```

Run firefox with
lower priority



```
nice -n -5  
make
```

Run make with
higher priority



```
nice -n 15  
python script.py
```

Run python script
with low priority

Nice Values :

- ✓ Range: -20 (highest priority) to 19 (lowest priority)
- ✓ Default value: 0
- ✓ Lower value = More priority
Higher value = Less priority



Tip

Use nice for background or non-critical tasks so that important processes get more CPU time.



FOLLOW
@devopswithanu



Small priority change,
Big performance impact!



Swipe
→

Day 14/20

8. ps Command

Display information about running processes.

What is ps?

The `ps` command shows a snapshot of the current running processes.



Syntax :

`ps [options]`

Commonly Used Options :

- `ps` → Current user processes
- `ps aux` → All running processes
- `ps -e` → All processes
- `ps -ef` → Full format listing
- `ps -u <user>` → Processes of specific user

Output Columns (common) :

`PID` → Process ID
`USER` → User of the process
`%CPU` → CPU usage
`%MEM` → Memory usage
`STAT` → Process state
`COMMAND` → Command name

ps gives you a quick view of what's running!

Examples :

Command	Example	Description
<code>ps</code>	<code>ps</code>	Show processes for current shell
<code>ps aux</code>	<code>ps aux</code>	Show all processes with details
<code>ps -e</code>	<code>ps -e</code>	Show all processes
<code>ps -u anurag</code>	<code>ps -u anurag</code>	Show processes of user 'anurag'

Sample Output (`ps aux`)

USER	PID	%CPU	%MEM	COMMAND
anu	1234	2.1	1.5	firefox
root	1	0.1	0.1	systemd
anu	5678	1.3	0.8	code
...				



FOLLOW
@devopswithanu



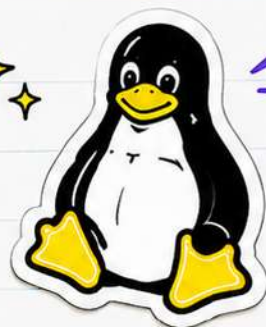
Pro Tip

Use `ps aux | grep <process_name>` to search specific processes. ✓

Swipe →
More coming! 😊

Day 14/20

9. top Command



Monitor system processes in real-time.

What is top?

The top command provides a dynamic real-time view of running processes and system resource usage.



Syntax: ♥

top



Key Information:

- ✓ System uptime & current time
- 👤 Number of users logged in
- 🔧 CPU usage (user, system, idle)
- 💾 Memory usage (total, used, free)
- 📊 List of running processes

Useful Keys:

P	→ Sort by CPU usage
M	→ Sort by Memory usage
k	→ Kill a process
q	→ Quit top
h	→ Show help

♥ FOLLOW
@devopswithanu



Use top to quickly check system health! ♥

Swipe →
Next command 😊

Day 14/20

10. You're Awesome!



You made it to the end of Part 5 (41-50)! 🎉

★ Quick Recap



We explored 10 powerful Linux commands.



These commands help manage and monitor your system efficiently.

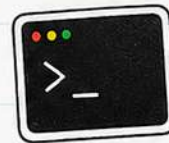


Practice them regularly to become a Linux pro! ❤️

Pro Tips 💡

- ✓ Use `man <command>` to read more about any command.
- ✓ Combine commands using `|`, `>`, `>>`
- ✓ Use `TAB` for auto-completion.
- ✓ Keep exploring and stay consistent! 💜

★ Every expert was once a beginner.
Keep learning, keep growing! 💙



See you in
Part 6 (51-60)! ↗️

❤️ FOLLOW
@devopswithanu

FOLLOW FOR MORE

Linux commands, tips, tricks
& DevOps content! ✨



★ ★ Thank you for learning with me! ❤️ ★ ★